

1.1.

$$\begin{aligned}\operatorname{Re}(\langle u, v \rangle) &= \frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) \quad \text{z definice Eukleidovské normy} \\ &= \frac{1}{4}(\langle u+v, u+v \rangle - \langle u-v, u-v \rangle) \quad \text{díky vlastnosti SL1 a SL2} \\ &= \frac{1}{4}(\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle - \langle v, v \rangle) \quad \text{díky vlastnosti SSS} \\ &= \frac{1}{4}(\langle u, v \rangle + \overline{\langle u, v \rangle} + \langle v, v \rangle + \langle u, v \rangle + \overline{\langle u, v \rangle} - \langle v, v \rangle) \\ &= \frac{1}{4}(2\operatorname{Re}(\langle u, v \rangle) + 2\operatorname{Re}(\langle u, v \rangle)) = \operatorname{Re}(\langle u, v \rangle)\end{aligned}$$

$$\begin{aligned}\operatorname{Im}(\langle u, v \rangle) &= \frac{1}{4}(\|u-iv\|^2 - \|u+iv\|^2) = \frac{1}{4}(\langle u-iv, u-iv \rangle - \langle u+iv, u+iv \rangle) \\ &= \frac{1}{4}(\langle u, u \rangle + \langle u, -iv \rangle + \langle -iv, u \rangle + \langle -iv, -iv \rangle - \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle - \langle iv, iv \rangle) \\ &= \frac{1}{4}(-i\langle u, v \rangle + \overline{\langle u, -iv \rangle} + \overline{(-i)} \cdot (-i) \langle v, v \rangle - i\langle u, v \rangle - \overline{\langle u, iv \rangle} - \overline{i} \cdot i \langle v, v \rangle) \\ &= \frac{1}{4}(-i\langle u, v \rangle + i\overline{\langle u, v \rangle} - i\langle u, v \rangle + i\overline{\langle u, v \rangle}) \\ &= \frac{-i}{4}(\langle u, v \rangle - \overline{\langle u, v \rangle} + \langle u, v \rangle - \overline{\langle u, v \rangle})\end{aligned}$$

$$\text{Necht' } \langle u, v \rangle = a+bi \quad \text{pak} \quad \langle u, v \rangle - \overline{\langle u, v \rangle} = a+bi - (a-bi) = 2bi = 2i\operatorname{Im}(\langle u, v \rangle)$$

$$= \frac{-i}{4} \cdot 4i\operatorname{Im}(\langle u, v \rangle) = \operatorname{Im}(\langle u, v \rangle)$$